FOOD DELIVERY MANAGEMENT SYSTEM – CASSINO 2025

Data Management Project Report

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1. INTRODUCTION

This project presents a comprehensive Food Delivery Management System designed for the city of Cassino tailored for integration with platforms like Glovo and Deliveroo. The system manages and analyzes food delivery operations, including customer orders, restaurant activities, driver assignments, and delivery performance.

The project integrates database design, SQL implementation, and data analysis using Python to transform raw transactional data into meaningful business insights. Cassino's diverse food environment, including pizzerias, sushi restaurants, kebab shops, and traditional eateries, provides a realistic and relevant context for the study.

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2. PROJECT OBJECTIVES

The main objectives of this project are:

1. To design a relational database for food delivery operations
2. To develop an Entity–Relationship (ER) model representing the system
3. To implement the database using SQL
4. To create analytical SQL queries for business evaluation
5. To perform data analysis using Python
6. To generate visualizations supporting managerial decision-making

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3. DATABASE DESIGN

3.1 Conceptual Model (ER Diagram)

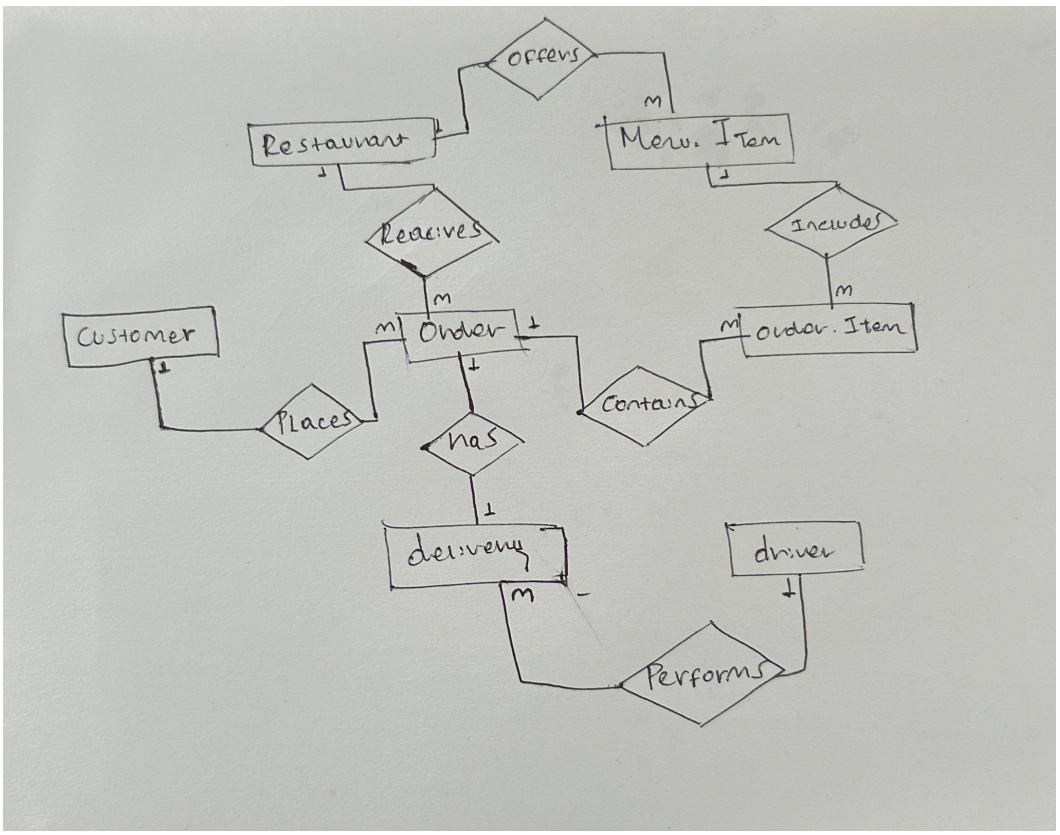

Figure 1: Entity–Relationship Diagram of the Food Delivery System

The system is composed of seven main entities:

- **Customer** – users who place orders
- **Restaurant** – food establishments registered on the platform
- **Driver** – delivery personnel
- **Order** – customer transactions
- **Menu_Item** – food items offered by restaurants
- **Order_Item** – items included in each order
- **Delivery** – delivery execution and feedback

Relationships:

- Customer (1) -- places -- (M) Order
- Restaurant (1) -- receives -- (M) Order
- Restaurant (1) -- offers -- (M) Menu_Item
- Order (1) -- contains -- (M) Order_Item
- Menu_Item (1) -- included in -- (M) Order_Item
- Order (1) -- has -- (1) Delivery
- Driver (1) -- performs -- (M) Delivery

3.2 Logical Model (Relational Schema)

The ER model is translated into the following relational schema:

1. **Customer** (`customer_id` PK, name, email, phone, address, registration_date)
2. **Restaurant** (`restaurant_id` PK, name, cuisine_type, address, delivery_fee)
3. **Driver** (`driver_id` PK, name, phone, vehicle_type, status)
4. **Order** (`order_id` PK, customer_id FK, restaurant_id FK, order_date, total_amount, status)
5. **Menu_Item** (`item_id` PK, restaurant_id FK, name, price, category, is_available)
6. **Order_Item** (`order_item_id` PK, order_id FK, item_id FK, quantity, unit_price)
7. **Delivery** (`delivery_id` PK, order_id FK, driver_id FK, pickup_time, delivered_time, rating)

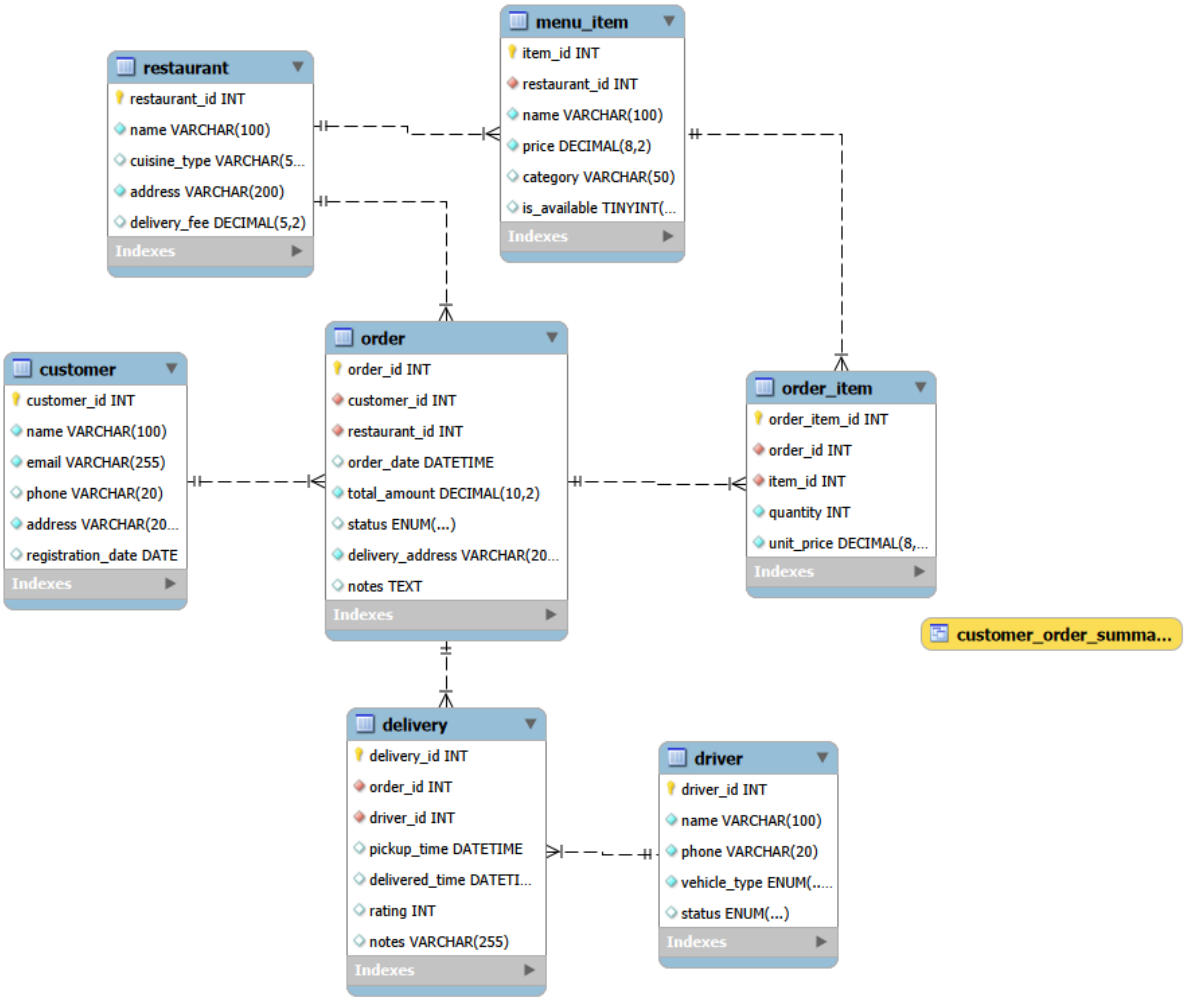

Figure 2: MySQL Workbench Relational Schema

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4. SQL IMPLEMENTATION

4.1 Database Creation

The database was implemented using MySQL. Tables were created with appropriate primary keys, foreign keys, and constraints to ensure data integrity and normalization.

4.2 Sample Data

The database contains realistic sample data representing food delivery activity in Cassino. The restaurants included in the dataset are real establishments operating in Cassino as of December 2025. This approach enhances the realism of the simulation and ensures that the analysis reflects plausible ordering patterns, customer behavior, and delivery operations within the local context.

Table	Records
Customer	14
Restaurant	10
Driver	8
Order	38
Menu_Item	62
Order_Item	68
Delivery	29

4.3 Key Queries

Analytical SQL queries were developed using JOIN, GROUP BY, aggregation functions, and window functions to analyze restaurant performance, revenue, customer behavior, and delivery efficiency.

Query 1 – Active Orders

Shows all orders that are still in progress and not yet delivered or cancelled.

Query 2 – Restaurant Performance
Compares restaurants based on total delivered orders, total revenue, and average order value.

Query 3 – Popular Menu Items
Identifies the most sold menu items and the revenue they generate.

Query 4 – Best Customers
Finds the most valuable customers based on number of orders and total spending.

Query 5 – Monthly Revenue Trend
Analyzes how orders and revenue change month by month.

Query 6 – Driver Performance
Evaluates drivers based on completed deliveries, average ratings, and delivery time.

Query 7 – Recent Order Details
Displays recent orders with customer, restaurant, items ordered, delivery time, and rating.

Query 8 – Ice Cream Seasonal Analysis
Analyzes ice cream sales to identify seasonal trends and peak months.

Query 9 – Busiest Hours Analysis
Identifies the busiest hours of the day based on delivered orders.

Query 10 – Cancellation Analysis
Analyzes cancelled orders and their reasons to understand service issues.

Query 11 – Customer Lifetime Value
Groups customers by value level and analyzes their contribution to total revenue.

Query 12 – Delivery Time by Vehicle Type
Compares delivery speed and ratings based on the driver's vehicle type.

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5. DATA ANALYSIS WITH PYTHON

5.1 Methodology

Data exported from the database was analyzed using Python in Google Colab with **pandas**, **matplotlib**, and **seaborn**. The workflow included:

1. **Data Loading** – Importing CSV files for orders, restaurants, drivers, and deliveries.

2. **Data Cleaning** – Handling missing values, formatting dates, and ensuring consistency.

3. **Feature Extraction** – Creating new columns for month, hour, weekday, and customer segments.

4. **Metric Calculation** – Computing total revenue, average order value, completion rate, peak hours, and active customers.

5. **Visualization** – Generating dashboards, bar charts, pie charts, and line plots to summarize key business metrics.

5.2 Results and Visualizations

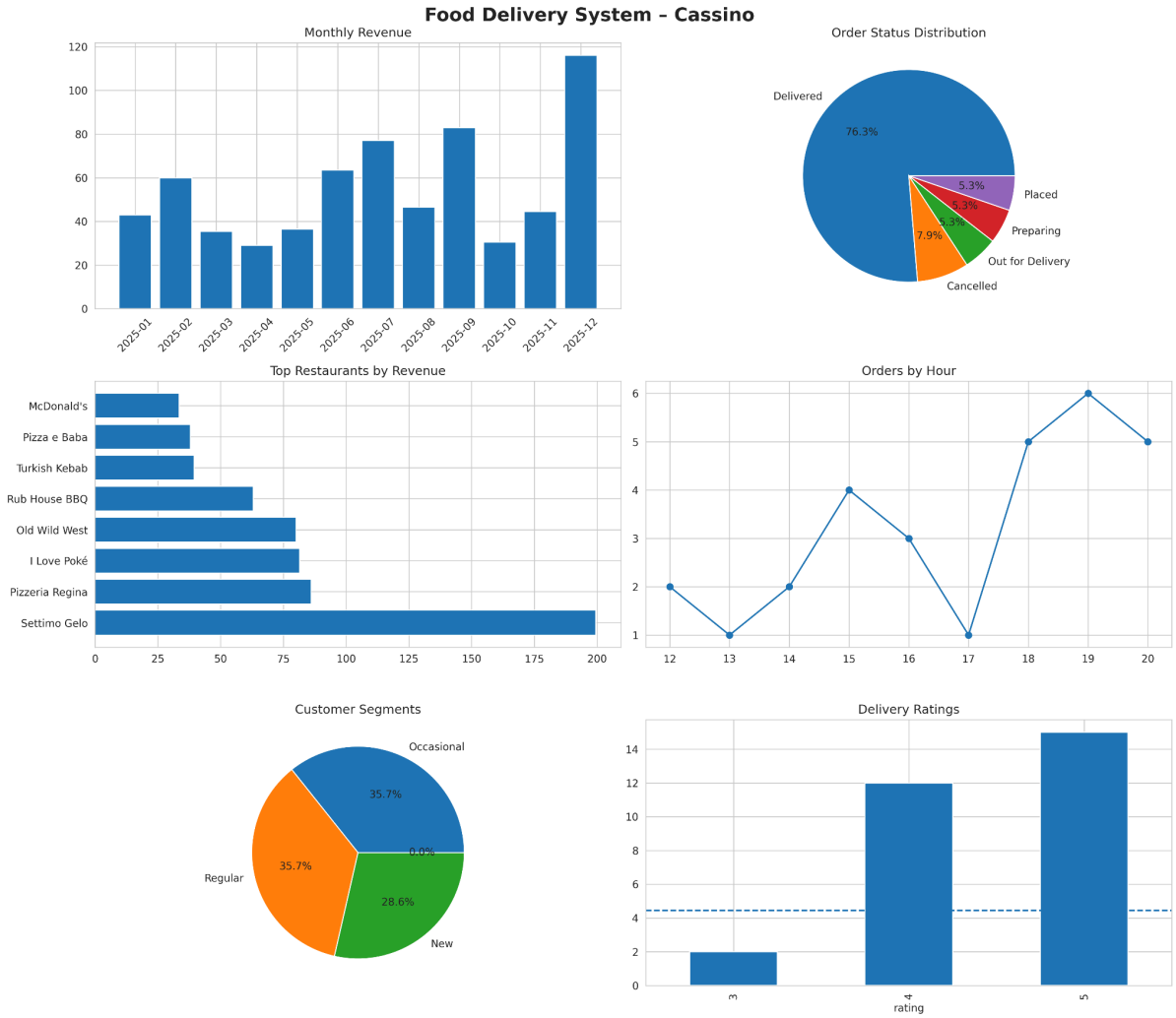

Figure 3: Food Delivery System Dashboard

The dashboard includes:

- Monthly revenue trend
- Order status distribution
- Top restaurants by revenue
- Hourly order demand
- Customer segmentation
- Delivery rating distribution

5.3 Key Findings

Metric	Value	Interpretation
Total Orders	38	Complete order dataset
Delivered Orders	29	Successful deliveries
Total Revenue	€665.00	Total income generated
Completion Rate	76.3%	Order success percentage
Peak Hour	19:00	Busiest ordering time
Best Month	December 2025	Highest revenue month
Active Customers	14	Unique customers served
Avg Order Value	€22.93	Average spending per order
Avg Delivery Rating	4.45/5	Customer satisfaction score

6. BUSINESS INSIGHTS AND RECOMMENDATIONS

6.1 Key Insights

- **Overall Performance:** The food delivery system shows stable and positive performance based on Python analysis of 2025 data.
- **Revenue:** Total revenue reached **€665.00**, with an average order value of **€22.93**.
- **Seasonality: December 2025** is the highest-revenue month, indicating strong holiday demand.
- **Order Volume:** A total of **38 orders** were processed, of which **29 were successfully delivered**.
- **Operational Efficiency:** The **76.3% completion rate** and **7.9% cancellation rate** indicate reliable operations.
- **Customer Base:** The system served **14 active customers**, with an average of **2.1 orders per customer**, suggesting repeat usage.
- **Service Quality:** An average delivery rating of **4.45/5** reflects high customer satisfaction.
- **Demand Patterns:** The **peak ordering hour is 20:00**, corresponding to evening demand.

- **Restaurant Network:** The platform integrates **10 partner restaurants**, with **Ice Cream** as the top-performing category.

Note: These results are based on my Python analysis of data extracted from the database. With a larger dataset, It is possible to draw more accurate and generalizable conclusions.

6.2 Strategic Recommendations

Marketing Optimization:

- Focus promotions during 18:00–21:00 peak hours
- Create holiday season campaigns (November–December)
- Target repeat customers with loyalty programs

Operational Improvements:

- Increase driver availability during peak hours
- Implement order confirmation system to reduce cancellations
- Optimize delivery routes based on historical patterns

Technology Enhancements:

- Simulate a delivery platform similar to Glovo for easier ordering
- Integrate real-time order tracking in the simulation
- Use predictive analytics to forecast demand and optimize driver allocation

7. CONCLUSION

The Food Delivery Management System for Cassino successfully demonstrates the integration of database design, SQL implementation, and data analysis. Key achievements include:

- Comprehensive database design with 7 properly normalized tables
- Realistic sample data with 161 records across all tables

- Twelve analytical SQL queries providing business intelligence
- Python-based data analysis with six insightful visualizations
- Data-driven recommendations for business improvement

The analysis shows a healthy business with €665.00 revenue, 76.3% completion rate, and an excellent 4.45/5 customer satisfaction score, providing a solid foundation for real-world implementation.

With a larger dataset covering more customers and a longer period, the results would reflect more accurate and statistically significant insights, enabling improved trend analysis, seasonal forecasting, and predictive modeling.

8. TOOLS AND TECHNOLOGIES USED

1. **Database Management:** MySQL 8.0 with MySQL Workbench
2. **Programming Language:** Python 3.9
3. **Data Analysis Libraries:** pandas, NumPy, matplotlib, seaborn
4. **Development Environment:** Google Colab
5. **Documentation:** Microsoft Word / Google Docs

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9. APPENDICES

Appendix A – SQL Script

Food Delivery Management System.sql

Appendix B – Python Analysis Code

food_delivery_system_analysis.py

Appendix C – Generated Outputs

- food_delivery_dashboard.png
- weekday_revenue.png
- business_analysis_summary.csv
- key_metrics.csv

Note: All appendix files are included in the submission folder.

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